

Java Programming 2

1. Which Java keyword is used for exception propagation (delegating exception handling to the calling method)?

- A) ``try``
- B) ``catch``
- C) ``throw``
- D) `**`throws`**`

Answer: D) ``throws``

2. What is the `**default capacity**` of an ``ArrayList`` when it is created with no arguments?

- A) 0
- B) `**10**`
- C) 16
- D) 100

Answer: B) 10

3. Which feature was introduced in Java 7 to simplify resource management in ``try-catch-finally`` blocks?

- A) Checked Exceptions
- B) Multi-catch blocks
- C) `**Try-with-Resources**`
- D) ``finally`` block removal

Answer: C) Try-with-Resources

4. What is the purpose of the `**`volatile`**` keyword in Java?

- A) To ensure that a variable is stored in the CPU cache.
- B) To prevent a variable from being serialized.
- C) `**To ensure that a variable's value is always read from the main memory and not from thread-local caches.**`
- D) To make a variable constant.

Answer: C) To ensure that a variable's value is always read from the main memory and not from thread-local caches.

5. Which class in Java is used to represent an arbitrary-precision decimal number?

- A) `Double`
- B) `Float`
- C) `BigDecimal`
- D) `BigFloat`

Answer: C) `BigDecimal`

6. In a `switch` statement, what happens if a `case` matches but there is no `break` statement?

- A) The program terminates.
- B) It causes a compile-time error.
- C) `Execution "falls through" to the next case label.`
- D) It automatically executes the `default` block.

Answer: C) Execution "falls through" to the next `case` label.

7. Which of the following is an example of an `Unchecked Exception`?

- A) `IOException`
- B) `SQLException`
- C) `ArithmeticException`
- D) `ClassNotFoundException`

Answer: C) `ArithmeticException`

8. Which standard annotation is used to tell the compiler that a method is intentionally replacing a method in its superclass or implemented interface?

- A) `@Deprecated`
- B) `@SuppressWarnings`
- C) `@Override`
- D) `@FunctionalInterface`

Answer: C) `@Override`

9. In Java, **method overriding** is an example of which type of polymorphism?

- A) Compile-time polymorphism
- B) **Runtime polymorphism (Dynamic Method Dispatch)**
- C) Static polymorphism
- D) Operator polymorphism

Answer: B) Runtime polymorphism (Dynamic Method Dispatch)

10. What is the process of converting a JSON or XML data structure into a Java object called?

- A) Serialization
- B) **Deserialization (or Marshalling)**
- C) Parsing
- D) Transpiling

Answer: B) Deserialization (or Marshalling)

11. Which data type is used to store boolean values?

- A) `int`
- B) `char`
- C) **`boolean`**
- D) `byte`

Answer: C) `boolean`

12. What is the purpose of the `System.exit(0);` method call?

- A) To close the current window.
- B) To throw an exception.
- C) **To terminate the currently running JVM with a status code.**
- D) To flush the output streams.

Answer: C) To terminate the currently running JVM with a status code.

13. What is the function of the `package` statement in a Java file?

- A) To define the name of the main class.
- B) To include external libraries.

C) **To declare the package to which the classes in the file belong.**

D) To restrict access to the file.

Answer: C) To declare the package to which the classes in the file belong.

14. What are the elements in a **Map** collection accessed by?

A) Index

B) **Key**

C) Hash Code

D) Iterator

Answer: B) Key

15. Which **Map** implementation maintains the insertion order of key-value pairs?

A) **HashMap**

B) **TreeMap**

C) **LinkedHashMap**

D) **WeakHashMap**

Answer: C) **LinkedHashMap**

16. Which method is used to retrieve the value associated with a specific key in a **HashMap**?

A) **find(key)**

B) **getValue(key)**

C) **get(key)**

D) **retrieve(key)**

Answer: C) **get(key)**

17. What is the minimum number of threads required to execute a Java program?

A) 0

B) **1 (the main thread)**

C) 2

D) The number of CPU cores

Answer: B) 1 (the main thread)

18. Which method is used to create an instance of a class using a default constructor via Java Reflection?

- A) `Class.forName()`
- B) `Constructor.newInstance()`
- C) `**Class.newInstance()` (deprecated) or `Constructor.newInstance()` `**`
- D) `Object.create()`

Answer: C) `Class.newInstance()` (deprecated) or `Constructor.newInstance()`

19. Which operator is used to calculate the remainder of a division?

- A) `/` (Division)
- B) `//` (Comment)
- C) `**%` (Modulo) `**`
- D) `&` (Bitwise AND)

Answer: C) `%` (Modulo)

20. What is the output of the following Java code?

<!-- end list -->

```
```java
String str = "Java";
str.concat(" Programming");
System.out.println(str);
```
```

- A) Java Programming
- B) `**Java**`
- C) Programming
- D) Compiler Error

Answer: B) Java

21. Which type of variable is shared among all instances of a class?

- A) Local variable
- B) Instance variable
- C) ****Static variable (Class variable)****
- D) Final variable

Answer: C) Static variable (Class variable)

22. What is the correct keyword to use when defining a method that returns no value?

- A) ``null``
- B) ``return``
- C) **`**`void`**`**
- D) ``empty``

Answer: C) ``void``

23. Which data structure in the Collections Framework provides LIFO (Last-In, First-Out) behavior?

- A) ``Queue``
- B) **`**`Stack`**`**
- C) ``Deque``
- D) ``ArrayList``

Answer: B) ``Stack``

24. Which ``List`` implementation is generally best for rapid random access (accessing elements by index)?

- A) ``LinkedList``
- B) ``Vector``
- C) **`**`ArrayList`**`**
- D) ``Stack``

Answer: C) ``ArrayList``

25. What is the purpose of the **`**`final`**`** keyword when applied to a method parameter?

- A) It must be initialized to 0.
- B) The parameter is passed by reference.
- C) **The parameter cannot be reassigned within the method.**
- D) The method cannot be overloaded.

Answer: C) The parameter cannot be reassigned within the method.

26. The **default** keyword (Java 8) in an interface is used to define:

- A) Static methods.
- B) **Methods with a concrete implementation.**
- C) Abstract methods.
- D) Private methods.

Answer: B) Methods with a concrete implementation.

27. Which class is the superclass for all stream classes in Java I/O for byte streams?

- A) `Reader`
- B) `Writer`
- C) **`InputStream`**
- D) `Stream`

Answer: C) `InputStream`

28. Which of the following is true about a **Constructor**?

- A) It must return a value.
- B) It can be declared as `abstract`.
- C) **It cannot be inherited.**
- D) It can be called explicitly like a regular method.

Answer: C) It cannot be inherited.

29. What is the term for the situation where an object is passed as an argument to a method, and the method's modification to the object is visible outside the method?

- A) Pass by Value
- B) **Pass by Reference (simulated through object references)**

C) Pass by Copy

D) Deep Copy

Answer: B) Pass by Reference (simulated through object references)

30. In Java, when you use a literal integer value like `10`, what is its default data type?

A) `long`

B) `short`

C) `int`

D) `byte`

Answer: C) `int`

31. How is a `varargs` (variable arguments) parameter declared in a method signature?

A) `(int... nums)`

B) `(int[] nums)`

C) `(varargs int nums)`

D) `(int nums*)`

Answer: A) `(int... nums)`

32. The `@Deprecated` annotation indicates that:

A) The element is a static final field.

B) `The element should no longer be used, often because a replacement exists.`

C) The element is only available in older Java versions.

D) The element is abstract.

Answer: B) The element should no longer be used, often because a replacement exists.

33. Which of the following is a key characteristic of the `Stream` API (Java 8)?

A) They are persistent data structures.

B) `They are lazily evaluated (intermediate operations are not executed until a terminal operation is invoked).`

C) They are fixed in size and cannot be modified.

D) They are primarily used for network communication.

Answer: B) They are lazily evaluated (intermediate operations are not executed until a terminal operation is invoked).

34. Which keyword is used to access the instance of the superclass from the subclass?

- A) `this`
- B) `parent`
- C) `super`
- D) `base`

Answer: C) `super`

35. What is the primary purpose of the `ExecutorService` interface in the `java.util.concurrent` package?

- A) To manage memory allocation.
- B) To provide a standardized way to submit and manage tasks for asynchronous execution (thread pooling).
- C) To handle I/O operations.
- D) To define the entry point of a program.

Answer: B) To provide a standardized way to submit and manage tasks for asynchronous execution (thread pooling).

36. Which of the following is a type-safe and preferred way to access elements of an `enum`?

- A) By direct integer value.
- B) By using a String constant.
- C) `Using the enum's constant name (e.g., Color.RED).`
- D) By reflection only.

Answer: C) Using the enum's constant name (e.g., `Color.RED`).

37. What happens when a static method in a superclass has the exact same signature as a static method in a subclass?

- A) Method Overriding
- B) Compile-time error
- C) `Method Hiding`
- D) Dynamic Dispatch

Answer: C) Method Hiding

38. What is the term for the memory area where objects are allocated in Java?

- A) Stack
- B) **Heap**
- C) Method Area
- D) Code Segment

Answer: B) Heap

39. The **@FunctionalInterface** annotation is used to designate an interface with:

- A) No methods.
- B) Multiple abstract methods.
- C) **Exactly one abstract method.**
- D) Only static methods.

Answer: C) Exactly one abstract method.

40. What is the behavior of the **notify()** method on an object?

- A) It wakes up all threads waiting on that object's monitor.
- B) It permanently releases the lock on the object.
- C) **It wakes up a single thread waiting on that object's monitor (if any).**
- D) It immediately transfers the lock to another thread.

Answer: C) It wakes up a single thread waiting on that object's monitor (if any).

41. Which class should be used for mutable, thread-safe character sequences?

- A) **String**
- B) **StringBuffer**
- C) **StringBuilder**
- D) **StringPool**

Answer: B) **StringBuffer**

42. How do you prevent a variable from being changed after it is initialized?

- A) Use the ``static`` keyword.
- B) ****Use the ``final`` keyword.****
- C) Declare it as ``private``.
- D) Use the ``const`` keyword.

Answer: B) Use the ``final`` keyword.

43. Which loop structure in Java is primarily designed for iterating over elements of arrays and collections (Java 5+)?

- A) ``for`` loop with index
- B) ``while`` loop
- C) ****Enhanced ``for`` loop (for-each loop)****
- D) ``do-while`` loop

Answer: C) Enhanced ``for`` loop (for-each loop)

44. Which exception is typically thrown by the JVM when the stack space is exhausted during excessive method recursion?

- A) ``OutOfMemoryError``
- B) ``ArithmeticException``
- C) ****``StackOverflowError``****
- D) ``SystemException``

Answer: C) ``StackOverflowError``

45. The ****``continue``**** statement in a loop:

- A) Immediately terminates the entire loop.
- B) ****Skips the current iteration and proceeds to the next one.****
- C) Restarts the loop from the beginning.
- D) Causes a run-time error.

Answer: B) Skips the current iteration and proceeds to the next one.

46. Which character is used to explicitly denote a ``long`` literal value?

- A) ``l`` (lowercase L) or ``L`` (uppercase L)
- B) ``d``

C) ``f``

D) ``i``

Answer: A) ``l`` (lowercase L) or ``L`` (uppercase L)

47. What is the purpose of the `System.gc()` method?

A) It immediately starts the Garbage Collector.

B) It allocates more memory to the heap.

C) `It suggests to the JVM to run the Garbage Collector, but does not guarantee execution.`

D) It cleans up all static variables.

Answer: C) It suggests to the JVM to run the Garbage Collector, but does not guarantee execution.

48. What is the type of initialization for a local variable in Java?

A) Zero-initialized

B) Null-initialized

C) Default-initialized

D) `Must be explicitly initialized before use.`

Answer: D) Must be explicitly initialized before use.

49. What does the term `instance initializer block` refer to?

A) Code inside the constructor.

B) `A non-static block of code inside a class, executed before the constructor when an object is created.`

C) A static block of code executed when the class is loaded.

D) The first line of the main method.

Answer: B) A non-static block of code inside a class, executed before the constructor when an object is created.

50. Which operator is used for `short-circuit OR` logic?

A) ``|``

B) ``^``

C) `||`

D) `!``

Answer: C) `|`

51. What is the maximum number of times a Java class can extend another class?

- A) Unlimited
- B) 2
- C) 5
- D) **1**

Answer: D) 1

52. What is the maximum number of interfaces a Java class can implement?

- A) 1
- B) 2
- C) 10
- D) **Unlimited**

Answer: D) Unlimited

53. What is the purpose of the **`assert`** keyword (disabled by default)?

- A) To enforce thread safety.
- B) To check for type compatibility at compile time.
- C) **To test an assumption about the program state, typically for debugging.**
- D) To declare an abstract method.

Answer: C) To test an assumption about the program state, typically for debugging.

54. Which `Map` implementation keeps its keys sorted in their natural order or by a custom `Comparator`?

- A) `HashMap`
- B) `LinkedHashMap`
- C) **`TreeMap`**
- D) `WeakHashMap`

Answer: C) `TreeMap`

55. The `Optional<T>` class (Java 8) is primarily used to:

- A) Make method arguments optional.
- B) Represent a value that may or may not be present, helping to avoid `NullPointerException`'s.
- C) Create thread-safe collections.
- D) Define a new data type.

Answer: B) Represent a value that may or may not be present, helping to avoid `NullPointerException`'s.

56. Which of the following is true about a `private` constructor?

- A) It prevents the class from being extended.
- B) It must be called using `super()`.
- C) It prevents the class from being instantiated from outside the class (often used in Singleton patterns).
- D) It is implicitly called by the JVM.

Answer: C) It prevents the class from being instantiated from outside the class (often used in Singleton patterns).

57. Which type of I/O stream is designed for reading raw 8-bit bytes?

- A) `Reader`
- B) `Writer`
- C) `InputStream`
- D) `PrintStream`

Answer: C) `InputStream`

58. What is a `Record` (Java 16+)?

- A) A complex class used for database interaction.
- B) A concise way to create data-only classes, automatically providing a constructor, accessors, `equals()`, `hashCode()`, and `toString()`.
- C) A specialized `List` implementation.
- D) A class that can only contain static members.

Answer: B) A concise way to create data-only classes, automatically providing a constructor, accessors, `equals()`, `hashCode()`, and `toString()`.

59. What is the term for a block of code that is executed only once when a class is loaded into the JVM?

- A) Constructor
- B) Instance Initializer Block
- C) Method
- D) ****Static Initializer Block****

Answer: D) Static Initializer Block

60. What is the main characteristic of a ****transient**** field?

- A) It is always initialized to `null`.
- B) ****It is excluded from the default serialization process.****
- C) It is only visible within its package.
- D) It can only be accessed by static methods.

Answer: B) It is excluded from the default serialization process.

61. Which exception should be thrown when a method receives an illegal or inappropriate argument?

- A) `IOException`
- B) ****IllegalArgumentException****
- C) `NullPointerException`
- D) `IllegalStateException`

Answer: B) `IllegalArgumentException`

62. What is the default access level if no modifier is specified for a class or its members?

- A) `public`
- B) `private`
- C) `protected`
- D) ****Package-private (default)****

Answer: D) Package-private (default)

63. What is the minimum version of Java required to use Lambda Expressions?

- A) Java 6
- B) Java 7
- C) ****Java 8****
- D) Java 11

Answer: C) Java 8

64. Which interface defines the contract for classes that can be compared to each other (natural ordering)?

- A) `Comparator`
- B) `Comparable`
- C) `Sortable`
- D) `Orderable`

Answer: B) `Comparable`

65. Which class in Java is used to read input from the console (standard input)?

- A) `BufferedReader`
- B) `Console`
- C) **`Scanner`**
- D) `FileReader`

Answer: C) `Scanner`

66. What is the output of the following Java code?

<!-- end list -->

```
```java
System.out.println(10 + 5 + "Java");
```
```

- A) 105Java
- B) ****15Java****

C) Java105

D) Compiler Error

Answer: B) 15Java

67. Which of the following is true regarding **finalize** methods (overridden from `Object`)?

A) They are guaranteed to be called before the object is garbage collected.

B) They are used to close resources reliably.

C) **They are rarely used in modern Java and can be called zero or one time by the garbage collector.**

D) They must throw an exception.

Answer: C) They are rarely used in modern Java and can be called zero or one time by the garbage collector.

68. What does the term **"Type Erasure"** refer to in Java Generics?

A) The removal of unused code by the JIT compiler.

B) **The process where generic type information is removed during compilation, resulting in raw types at runtime.**

C) The use of the `var` keyword.

D) The dynamic nature of method overriding.

Answer: B) The process where generic type information is removed during compilation, resulting in raw types at runtime.

69. Which method of the `Thread` class is used to allow a thread to temporarily halt its execution and allow other threads to run?

A) `sleep()`

B) **`yield()`**

C) `join()`

D) `wait()`

Answer: B) `yield()`

70. Which class is used to wrap a primitive type `int` into an object?

A) `Int`

B) **`Integer`**

C) `Number`

D) `ObjectInt`

Answer: B) `Integer`

71. What is the process of extracting a primitive value from its corresponding wrapper object called?

A) Autoboxing

B) **Unboxing**

C) Casting

D) Narrowing

Answer: B) Unboxing

72. What is the main purpose of the `String.intern()` method?

A) To convert the string to a primitive array.

B) To make the string immutable.

C) **To place the string in the JVM's String constant pool, often returning a canonical reference.**

D) To serialize the string object.

Answer: C) To place the string in the JVM's String constant pool, often returning a canonical reference.

73. The `"is-a"` relationship is typically implemented using:

A) Association

B) Composition

C) Aggregation

D) **Inheritance or Interface Implementation**

Answer: D) Inheritance or Interface Implementation

74. The `"has-a"` relationship is typically implemented using:

A) Interface Implementation

B) **Composition or Aggregation**

C) Method Overriding

D) Static methods

Answer: B) Composition or Aggregation

75. Which operator is used for **Bitwise XOR**?

- A) `&`
- B) `|`
- C) `^`
- D) `~`

Answer: C) `^`

76. What does the `java.nio` package (NIO - New I/O) primarily provide?

- A) Legacy file handling mechanisms.
- B) **Non-blocking I/O operations and buffer-oriented data handling.**
- C) Database connectivity.
- D) Remote Method Invocation (RMI).

Answer: B) Non-blocking I/O operations and buffer-oriented data handling.

77. What is the effect of making a class method **final**?

- A) It can only be called by the class itself.
- B) **It cannot be overridden by subclasses.**
- C) It cannot access instance variables.
- D) It must return a `void` value.

Answer: B) It cannot be overridden by subclasses.

78. When a class implements an interface, it must:

- A) Extend the interface.
- B) **Provide concrete implementation for all abstract methods defined in the interface (unless the class is abstract).**
- C) Define a static initializer block.
- D) Declare all its own methods as `final`.

Answer: B) Provide concrete implementation for all abstract methods defined in the interface (unless the class is abstract).

79. What is the purpose of the `instanceof` operator in a conditional statement?

- A) To check if two objects are equal.
- B) **To check if an object is of a certain type (class or interface).**
- C) To determine the size of an object.
- D) To compare the hash codes of two objects.

Answer: B) To check if an object is of a certain type (class or interface).

80. In Java 8, which method is added to the `Collection` interface to enable bulk operations using a predicate?

- A) `removeElement()`
- B) **`removeIf()`**
- C) `bulkDelete()`
- D) `filterAndRemove()`

Answer: B) `removeIf()`

81. What is the primary use of the `public static final` modifiers in an `Interface` field?

- A) To enable serialization.
- B) To define instance variables.
- C) **To define constants (all interface fields are implicitly these modifiers).**
- D) To define static methods.

Answer: C) To define constants (all interface fields are implicitly these modifiers).

82. Which of the following is true about `Anonymous Inner Classes`?

- A) They can have multiple constructors.
- B) They must be declared as `static`.
- C) **They have no name and are instantiated and defined in a single expression.**
- D) They cannot access local variables from the enclosing scope.

Answer: C) They have no name and are instantiated and defined in a single expression.

83. What is the correct way to declare and initialize a variable of type `float`?

- A) `float f = 10.5;`

B) `**float f = 10.5f;**`

C) `float f = (float)10.5;`

D) Both B and C are correct.

Answer: D) Both B and C are correct. (A is incorrect because `10.5` is a `double` literal by default).

84. What is the order of execution when an object is created?

A) Instance block -> Constructor -> Static block.

B) Static block -> Instance block -> Main method.

C) `**Static block (when class is loaded) -> Instance block -> Constructor.**`

D) Constructor -> Instance block -> Static block.

Answer: C) Static block (when class is loaded) -> Instance block -> Constructor.

85. Which is the preferred class for reading binary data from a stream?

A) `FileReader`

B) `BufferedReader`

C) `**FileInputStream**`

D) `DataInputStream`

Answer: C) `FileInputStream`

86. What is the purpose of the `**@SuppressWarnings**` annotation?

A) To disable all compiler warnings globally.

B) `**To suppress specific warnings reported by the compiler for an annotated element.**`

C) To ensure the code is thread-safe.

D) To make all methods `private`.

Answer: B) To suppress specific warnings reported by the compiler for an annotated element.

87. Which feature of Java prevents one class from accidentally modifying the fields of another class?

A) Inheritance

B) `**Encapsulation (through access modifiers)**`

C) Polymorphism

D) Abstraction

Answer: B) Encapsulation (through access modifiers)

88. The `try-catch-finally` block structure handles exceptions by:

A) Executing the `try` block and then the `catch` block.

B) `Attempting the try block, executing catch if an exception occurs, and always executing finally.`

C) Executing `catch` first, then `try`.

D) Throwing an exception if no `finally` block exists.

Answer: B) Attempting the `try` block, executing `catch` if an exception occurs, and always executing `finally`.

89. Which class provides a number of utility methods for arrays, including binary search, equality checking, and range filling?

A) `java.util.Collections`

B) `java.util.Arrays`

C) `java.lang.Array`

D) `java.util.List`

Answer: B) `java.util.Arrays`

90. What is the correct way to handle multiple exceptions in a single `catch` block (Java 7+)?

A) `catch (Exception1, Exception2 e)`

B) `catch (Exception1 | Exception2 e)`

C) `catch (Exception1 | Exception2 e)`

D) `catch (Exception1 & Exception2 e)`

Answer: B) `catch (Exception1 | Exception2 e)` or `catch (Exception1 | Exception2 e)`
(Symbolic notation is key)

91. What does the `java.math.Math.random()` method return?

A) A positive `long` integer.

B) `A double value between 0.0 (inclusive) and 1.0 (exclusive).`

C) A positive `int` integer.

D) A random object.

Answer: B) A `double` value between 0.0 (inclusive) and 1.0 (exclusive).

92. Which package contains classes for defining custom Annotations?

- A) `java.lang`
- B) `java.util`
- C) `\*\*java.lang.annotation\*\*`
- D) `java.reflect`

Answer: C) `java.lang.annotation`

93. What is the primary difference between `Callable<V>` and `Runnable`?

- A) `Runnable` is thread-safe; `Callable` is not.
- B) `Callable` can return a value and throw checked exceptions; `Runnable` cannot.
- C) `Callable` is faster than `Runnable`.
- D) `Runnable` is used for single-threaded applications.

Answer: B) `Callable` can return a value and throw checked exceptions; `Runnable` cannot.

94. Which class is used to retrieve the result of an asynchronous computation submitted to an `ExecutorService`?

- A) `Thread`
- B) `Runnable`
- C) `\*\*Future\*\*`
- D) `Promise`

Answer: C) `Future`

95. What is the output of the following Java code?

<!-- end list -->

```
```java
int x = 5;
if (x > 5)
```

```
 System.out.println("A");
else if (x == 5)
 System.out.println("B");
else
 System.out.println("C");
...
```

A) A

B) **\*\*B\*\***

C) C

D) A B C

Answer: B) B

96. Which method is used to compare two strings lexicographically, ignoring case differences?

A) ``equals()``

B) ``compare()``

C) **`**`equalsIgnoreCase()`**`**

D) ``compareToIgnoreCase()``

Answer: D) ``compareToIgnoreCase()``

97. What is the term for defining multiple constructors in a single class with different parameter lists?

A) Method Overriding

B) **`**Constructor Overloading**`**

C) Constructor Hiding

D) Constructor Inheritance

Answer: B) Constructor Overloading

98. Which of the following is an example of an **`**Autoboxing**`** conversion?

A) ``Integer i = 100;``

B) ``int j = new Integer(50);``

C) ``String s = "Hello";``



D) `Object o = new Integer(10);`

Answer: A) `Integer i = 100;`

99. The `switch` statement in Java can use which data types for its control expression (in modern Java)?

A) `boolean`, `double`, `float`

B) `long`, `int`, `short`

C) `byte`, `short`, `char`, `int`, `String`, `enum`, and wrapper classes for these primitives.

D) Only `int` and `String`.

Answer: C) `byte`, `short`, `char`, `int`, `String`, `enum`, and wrapper classes for these primitives.

100. When using the `var` keyword for local variable type inference (Java 10+), which of the following is required?

A) The variable must be a field in a class.

B) The variable must be declared as `final`.

C) The variable must be initialized immediately with a value.

D) The variable must be an array type.

Answer: C) The variable must be initialized immediately with a value.